

BURKAN LOCAL AI

MICROSOFT SUMMER SCHOOL PROJECT
PRESENTATION

- YUŞUF BURKAN
- BOĞAZIÇI UNIVERSITY

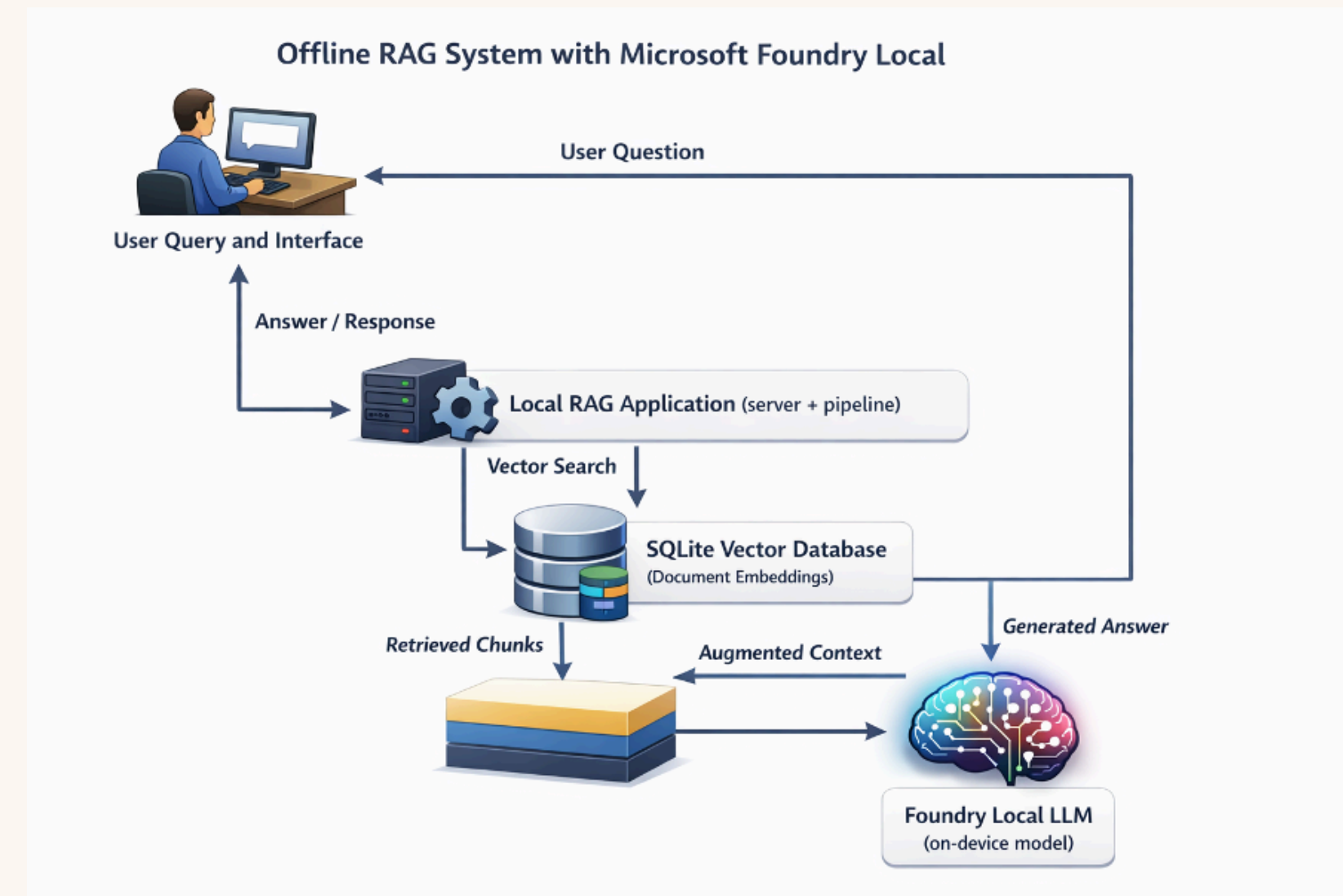


WHY DID I BUILD THIS PROJECT?

- CHATGPT CANNOT ACCESS PRIVATE PDFS.
- LLMS MAY HALLUCINATE.
- INTERNET MAY NOT ALWAYS BE AVAILABLE.
- NEED FOR A SECURE LOCAL ASSISTANT.

EXPECTED BENEFITS

- ✓ PRIVACY
- ✓ RELIABILITY
- ✓ OFFLINE ACCESS
- ✓ FASTER RETRIEVAL



PRIMARY GOALS

- LOCAL AI
RUNS ENTIRELY ON THE LOCAL MACHINE.
- OFFLINE
NO INTERNET REQUIRED.
- SEMANTIC SEARCH
MEANING-BASED DOCUMENT RETRIEVAL.
- SOURCE ATTRIBUTION
SHOWS DOCUMENT REFERENCES.
- REDUCE HALLUCINATIONS
ANSWERS ONLY FROM RETRIEVED CONTEXT.



PROJECT STRUCTURE

LOCALRAG

__pycache__

database.cpython-314.pyc

rag.cpython-314.pyc

.venv

docs

0 - intro.pdf

1 - money creation.pdf

knowledge.txt

Lecture 1 Introduction to Envir...

Lecture 2 Introduction to Ecol...

Lecture 3 Populations and Co...

Lecture 4 Ecosystems.pdf

.env

.gitignore

database.db

database.py

ingest.py

main.py

rag.py

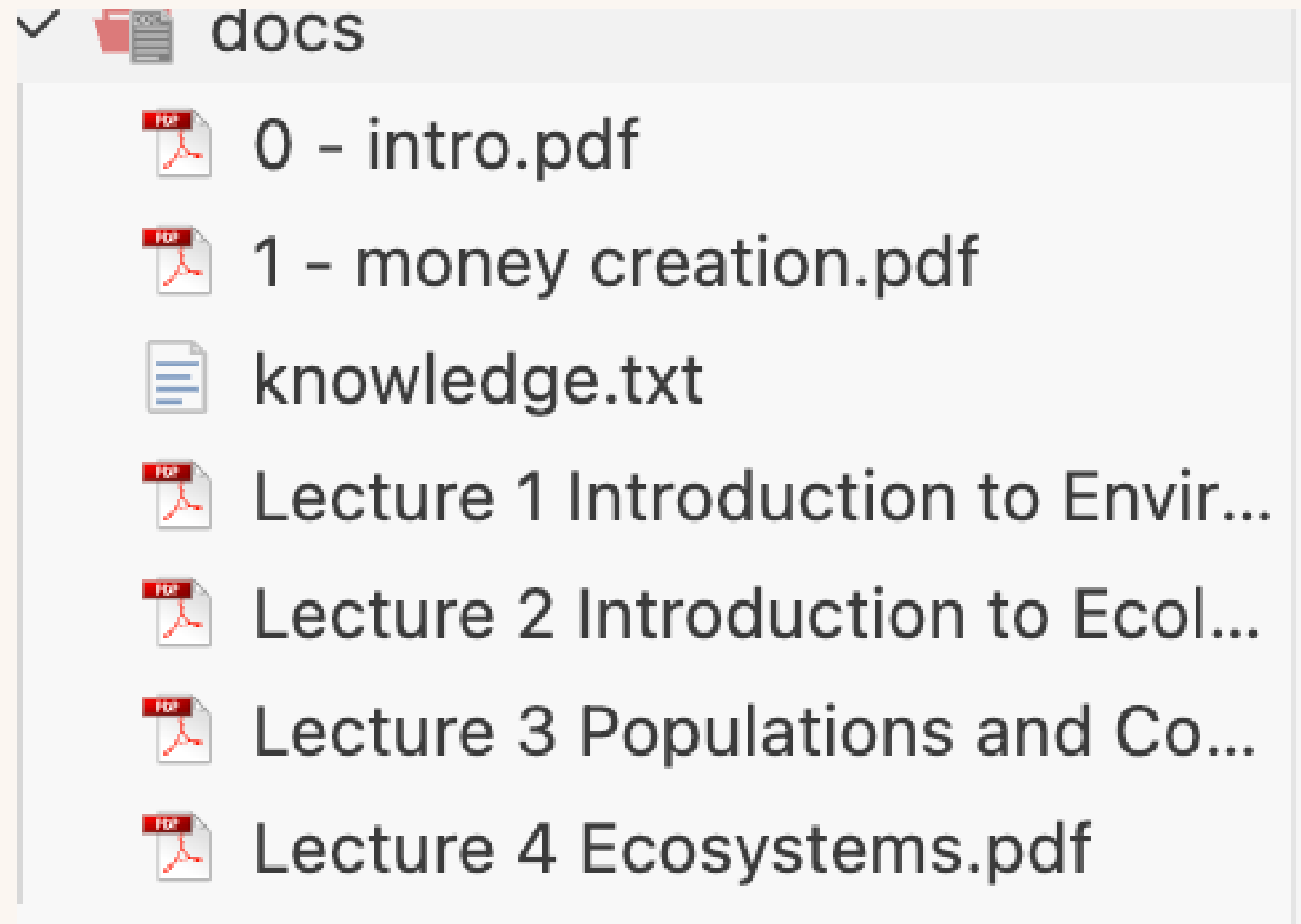
README.md

requirements.txt

test_foundry.py

File	Purpose
docs	PDF knowledge base
database.db	Stores chunks & embeddings
ingest.py	Document ingestion
rag.py	Retrieval engine
main.py	CLI application

KNOWLEDGE BASE



- KNOWLEDGE SOURCES
- LECTURE NOTES
- ENVIRONMENTAL SCIENCE PDFS
- TXT DOCUMENTS

STATISTICS

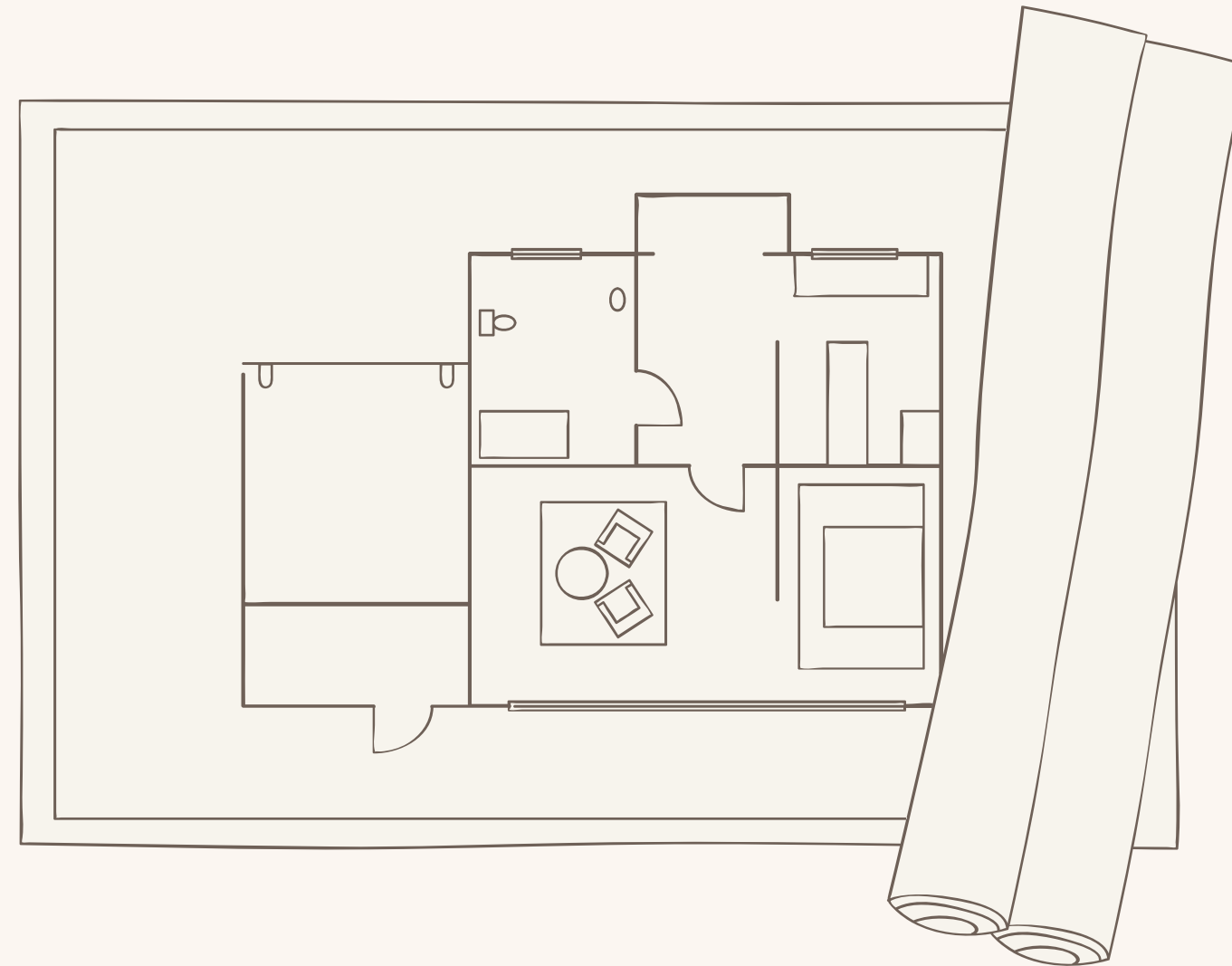
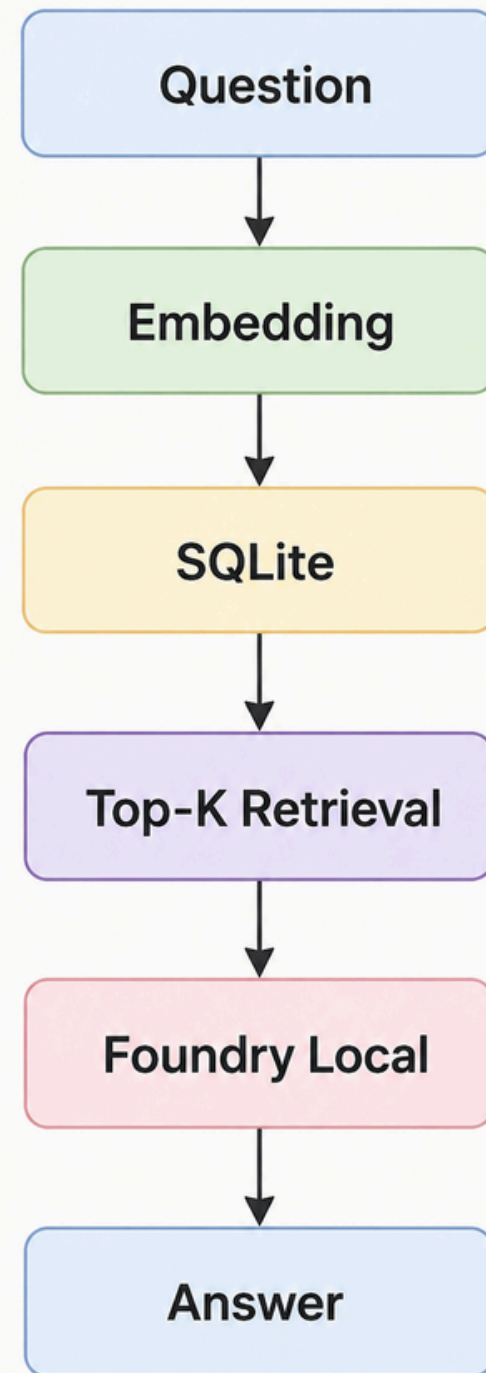
6 PDF FILES

1 TXT DOCUMENT

100+ DOCUMENT CHUNKS

LOCAL STORAGE

ARCHITECTURE



EACH STAGE CONTRIBUTES TO IMPROVING RETRIEVAL ACCURACY WHILE REDUCING HALLUCINATIONS.

PDF PROCESSING PIPELINE

PDF



TEXT EXTRACTION



CHUNKING



EMBEDDING GENERATION



SQLITE STORAGE

```
def read_txt(file_path: Path) -> str:
    return file_path.read_text(encoding="utf-8")

def read_pdf(file_path: Path) -> str:
    reader = PdfReader(file_path)
    pages = []

    for page_number, page in enumerate(reader.pages, start=1):
        page_text = page.extract_text()

        if page_text:
            pages.append(
                f"[Page {page_number}]\n{page_text.strip()}"
            )

    return "\n\n".join(pages)
```

DOCUMENTS ARE PROCESSED ONLY ONCE DURING INGESTION.

RETRIEVAL PROCESS

```
results = search_documents(question, top_k=3)

print("\n" + "=" * 55)
print("📖 En alakalı bilgiler")
print("=" * 55)

for index, result in enumerate(results, start=1):
    print(f"\n#{index}")
    print(f"📄 Source: {result['source']}")
    print(f"🧩 Chunk ID: {result['chunk_id']}")
    print(f"📈 Similarity Score: {result['score']:.4f}")
    print(f"\n{result['content']}")
```

- RETRIEVAL OUTPUT
- SOURCE
- CHUNK ID
- SIMILARITY SCORE
- RETRIEVED TEXT

PROMPT ENGINEERING

- Do not invent information.
- Use only information contained in the context.
- If the answer is not contained in the context, say exactly:
I could not find the answer in the local knowledge base.
- Give a short and clear answer.
- Answer in the same language as the user's question.
- Cite only the sources actually used.
- At the end, mention the relevant source numbers in this format:
Sources: [1], [2]

PROMPT RULES

- ✓ NO HALLUCINATIONS
- ✓ CONTEXT ONLY
- ✓ SAME LANGUAGE
- ✓ SOURCE CITATION
- ✓ SHORT ANSWERS

DATABASE

- STORED INFORMATION
- DOCUMENT CHUNKS
- EMBEDDINGS
- SOURCE METADATA
- CHUNK INDEX

```
SQLITE> SELECT ID, SOURCE, CHUNK_INDEX
        FROM DOCUMENTS
        LIMIT 5;
```

```
15 | DOCS/0 - INTRO.PDF | 1
16 | DOCS/0 - INTRO.PDF | 2
17 | DOCS/0 - INTRO.PDF | 3
18 | DOCS/0 - INTRO.PDF | 4
19 | DOCS/0 - INTRO.PDF | 5
```

SQLite Database Design		
Column		Description
 id		Unique chunk ID
 content		Text extracted from PDFs
[0.1, 0.2, ...] embedding		Vector representation
 source		Original PDF document
 chunk_index		Position of the chunk

DEMO

- DEMONSTRATED FEATURES
- SEMANTIC RETRIEVAL
- SOURCE ATTRIBUTION
- SIMILARITY SCORES
- LOCAL RESPONSE GENERATION



👤 WHAT IS YOUR NAME? BURKAN

👋 WELCOME, BURKAN!
YOU CAN ASK QUESTIONS BASED ON THE LOCAL KNOWLEDGE BASE.
TYPE 'EXIT' TO CLOSE THE APPLICATION.

💬 BURKAN, ASK A QUESTION: WHAT IS AN ECOSYSTEM?

=====

📖 RETRIEVED SOURCES

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#1

📄 SOURCE: DOCS/LECTURE 4 ECOSYSTEMS.PDF

🌱 CHUNK ID: 96

📈 SIMILARITY SCORE: 0.8033

📖 PREVIEW: [PAGE 1] ES C 301.0 3 THE ENVIRONMENTAL DIMENSION ECOSYSTEMS ASSOC. PROF. DR. BAŞAK GÜVEN [PAGE 2] AN ECOSYSTEM IS A DYNAMIC COMPLEX OF PLANT, ANIMAL AND MICROORGANISM COMMUNITIES ...

#2

📄 SOURCE: DOCS/LECTURE 4 ECOSYSTEMS.PDF

🌱 CHUNK ID: 97

📈 SIMILARITY SCORE: 0.6961

📖 PREVIEW: NUTRIENTS ❖ ABSORBING AND DETOXIFYING POLLUTANTS ❖ PROVIDING BEAUTY, INSPIRATION AND RECREATION [HTTPS://WWW.WHO.INT/GLOBALCHANGE/ECOSYSTEMS/EN/](https://www.who.int/globalchange/ecosystems/en/) ECOSYSTEM SERVICES ECOSYSTEM GOODS [...]

#3

📄 SOURCE: DOCS/LECTURE 3 POPULATIONS AND COMMUNITIES.PDF

🌱 CHUNK ID: 86

📈 SIMILARITY SCORE: 0.6570

📖 PREVIEW: E IN THE COMMUNITY AND HOW IT INTERACTS WITH THE ENVIRONMENT: I.E. HOW IT OBTAINS FOOD, MATES AND PROTECTION FROM PREDATORS. [PAGE 15] HABITAT NICHE: THE PLACE WHERE AN ORGANISM LI...

CHALLENGES

- **CHUNK SIZE**
CHOOSING AN OPTIMAL CHUNK LENGTH.
- **SIMILARITY THRESHOLD**
BALANCING RECALL AND PRECISION.
- **PROMPT DESIGN**
REDUCING HALLUCINATIONS.
- **PERFORMANCE**
FAST LOCAL RETRIEVAL.



THANK YOU!

[GITHUB.COM/YSFBRKAN/BURKAN-LOCAL-RAG](https://github.com/ysfbrkan/burkan-local-rag)